

Course Overview

- It introduces students to the fundamentals of information techn, computer use and essential software tool required in statistical work.

Role of IT in the society

- IT in the society is used in:
 - * Communication - Email, social media.
 - * Business - Accounting, data processing
 - * Education - E-learning, research.
 - * Health - Patient records, diagnosis
- It makes work easier and faster.

Information Technology (IT) - Refers to the use of computers and related devices to collect, process and communicate information

Computer - an electronic machine that accepts data (input), process it, and produce meaningful results (output)

History of Computing

- It is divided into five generations.

a) 1st Generation (Vacuum tubes)

- Computers used vacuum tubes for circuitry

Characteristics

- a) They were very large.
- b) They generated alot of heat.
- c) Slow and un-reliable.

b) 2nd Generation (Transistors)

- Transistors replaced vacuum tubes.

X'tics:

- a) They were smaller than Vacuum tubes.
- b) Beginning of stored-program computers
- c) They were cheaper and more reliable.

3rd (Integrated Circuits):

- Introduction of Integrated Circuits (Many transistor on one chip)

X'tics:

- * Much smaller and more powerful machines.
- * Use of keyboard and Monitors began.

4th (Microprocessors):

- Microprocessors invented. Entire CPU on a single chip.
- They led to personal computers (PCs)
- They led to developments of laptops and modern systems.

5th (Present and future) - Artificial Intelligence

- Based on AI, machine learning and parallel processing.
- Natural language processing, robotics

(2) Fundamentals of Computer Operations:

- A computer performs tasks through a basic cycle called the Input -> Processing -> Output -> (Storage) cycle
- Computer Operations are the basic activities a computer performs to convert data into useful information.
- Input - Process - Output cycle (IPO) cycle
- The IPO cycle describes how a computer works to turn raw data into useful information.

(a) Input:

This is when data or instructions are entered into a computer.

Examples:

- * Typing on a keyboard * Scanning a document
- * Clicking a mouse * Speaking into microphone

1) Processing

- The CPU (Central processing Unit) processes the input data
- The CPU has
 - a) ALU - Performs calculation and logical decisions.
 - b) Control unit - Directs all activities
 - c) Registers - Fast memory areas.

2) Output

This is when processed data is displayed as information.

Examples:

- * Displayed on Monitor. * Sound from speaker
- * Printed document. * Saved file.

3) Introduction to Operating System

- An operating system is the main software that controls the computer's hardware, manages programs and allows the user to interact with the computer.
- It acts as a bridge between the user and the computer.

Functions:

- User interface - Provides a way to interact with the computer (commands)
- File management - Create, delete, saves and organizes files and folders.
- Memory management - Controls how applications use RAM to run smoothly.
- Process Management - Manages multi-tasking.
- Security - Provides password, user account permissions.

4) Computer Hardware Basics:

- Computer hardware refers to the physical, touchable parts of the computer system.
Central Processing Unit (CPU)
The CPU is the brain of the computer. It performs all calculations and controls all operations.

Components of the CPU

- a) Arithmetic & Logic Unit (ALU) - Performs arithmetic operations.
- b) Control Unit (CU) - Directs all activities inside the computer.
- c) Registers - Very fast small memory locations inside CPU for temporary data.

Functions of the CPU

- Processing data.
- Executing instructions.
- Controlling hardware.
- Managing data flow.

Memory (Primary Storage)

Memory stores data and instructions temporarily or permanently.

a) Random Access Memory (RAM)

- Temporary/Volatile memory.
- Stores data and programs currently in use.
- Content is lost when power is switched off.

b) Read Only Memory (ROM)

- Permanent/Non-volatile memory.
- Stores start up instructions.
- Cannot be easily changed.

c) Cache Memory

- High speed memory between RAM and CPU.
- Speeds up processing.

Input devices

Used to enter data into the computer.

Examples:

- Keyboard - Scanner
- Mouse - Camera

Output devices

Used to display or produce information.

Examples:

- Monitor - Speakers
- Printer - Projector

Storage devices (Secondary storage)

- Used to store data permanently

Types:

- Hard disk drive (HDD) - Magnetic storage.
- Solid state drive (SSD) - Faster, no moving parts.
- Flash drive
- Memory card
- Optical disks: CD, DVD

Importance

- To store files permanently.
- Back up of data.
- Large capacity compared to RAM.

Computer Software Basics

- Computer software refers to the programs and instructions that tell the hardware what to do. It is divided into:

- * System software
- * Application software

a) System Software

- It controls and manages the hardware of the computer.
- It helps the computer function properly and provides a platform for application software to run.

Types of system software:

- Operating System (Windows, MacOS)
- Utility programs (Anti-virus, file manager)
- Device drivers (Printer driver, display driver)

Functions

- Manages hardware
- Loads and runs programs
- Manages memory and files

b) Application Software

- Allows the user to perform specific tasks.

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Two translating systems:

A. Compiler

- Translates the entire program all at once.
- Produces a separate machine code file.
- Faster execution.
- Examples: C, C++, Java.

B. Interpreter

- Translates the program line by line.
- Slower execution.
- Stops when an error is found.
- Examples: Python, JavaScript.

Types of Application Software

- * General-purpose – (Excel, MS Word, PowerPoint)
- * Specialized – (Accounting software)
- * Custom-made – (School management systems)
- * Web applications – (Browsers, email apps)

Operating Systems

- This is the main system software that manages:

- * Hardware devices
- * Input/output operations
- * Files and folders
- * Programs and memory
- * Users and security

Functions

- Provides a user interface (desktop)
- Manages memory and processes
- Controls hardware through drivers
- Handles files (create, delete, rename)

Compiling Systems

- Programming languages are written in human-readable form.
- Before the computer can run the program it must be translated into machine language.

Hand-On Demo ; Software installation

Steps in software installation

1) Obtain the software

-> From CD, USB, or downloaded setup file.

2) Run the Installer

-> Double-click the .exe (windows) or .pkg (Mac) file.

3) Follow the Installation Wizard

-> Accept license agreement.

-> Choose installation folder.

-> Select installation type (Typical / Custom)

4) Copying or extracting files

-> System copies program files to the hard disk.

-> Installer configures settings and registers the program.

5) Finishing Installation

-> Click finish when completed.

-> Restart PC if required

6) Test the installation

-> Open the program if it runs.

-> Check short cuts on desktop/start menu.

5) Data and Data Files

Concept of Data:

- Data refers to raw facts and figures that have no meaning on their own.

- Information is processed and meaningful data.

* Bits and Bytes

a) Bit (Binary digit)

- This is the smallest unit of data in a computer.

- Can only be zero or 1.

b) Byte

- This is a group of 8 bits.

- Represents a single character.

Records are stored one after another in a fixed order.

Atics:

- Data is arranged sequentially
- You must load the file from the beginning to the point you want
- Efficient for batch processing.

Advantages:

- > Simple to create.
- > Good for large ordered data.
- > Fast when reading all records.

Disadvantages:

- > Slow when searching randomly
- > Hard to update or delete individual record.

Other Units:

- Kilobyte (KB): 1024 bytes.
- Megabyte (MB): 1024 KB
- Gigabyte (GB): 1024 MB
- Terabyte (TB): 1024 GB.

Data types:

They specify the nature of data stored in a computer.

They include:

- Numeric: integers, decimals.
- Character: letters or symbols.
- String (Text): words.
- Boolean: True or false.
- Date/Time.

Data files:

A data file is a collection of related data stored under one name.

Example:

- > student-records.txt
- > Sals2023.xls

Types of data files:

- a) Sequential data files
- b) Random data files
- c) Sequential data files

3) Clusters

- A cluster (also called allocation unit) is a group of sectors.
- OS allocates disk space in clusters, not sectors.
- A file uses one or more clusters.

4) Surfaces

- A disk has multiple platters, and each platter has 2 surfaces:
 - Top surface
 - Bottom surface
- Data is stored on all surfaces, increasing capacity.

1) Random Data Files

- Records are stored using addresses so they can be accessed directly without reading earlier records.

Advantages:

- Fast searching
- Easy to update specific records
- Supports real-time processing

Disadvantages:

- More complex to design
- Requires more storage

6) Disk Storage Fundamentals

- Disk storage refers to the method and physical parts used to store data permanently on devices such as hard disks, SSDs, and flash drives.

1) Tracks

- A track is a circular path on the surface of a disk where data is recorded.
- Each track is divided into smaller parts called sectors.
- Tracks are centered from the centre outward.

2) Sectors

- A sector is the smallest physical storage unit on a disk.
- Normally stores 512 bytes or 4096 bytes.
- Many sectors make a track.
- Data is read/written one sector at a time.

- Cheap
- Good for distributing media/software

Disadvantages:

- Slow read/write speed
- Easily scratched

3. Solid-State Storage

- Uses flash memory (no moving parts).

Examples:

- SSD
- Flash drives
- Memory cards

Advantages:

- Very fast
- Durable
- Low power use

Disadvantages:

- More expensive than HDD

Types of Disk Storage

1. Magnetic Storage

- Uses magnetic patterns to store data.

Examples:

- Hard Disk Drive (HDD)
- Magnetic tapes

Advantages:

- Large capacity
- Cheap per GB

Disadvantages:

- Slower than SSDs
- Has moving parts (can fail)

2. Optical Storage

- Uses laser technology to read/write data.

Examples:

- CD (700MB)
- DVD (4.7GB – 8.5GB)
- Blu-ray (25GB – 50GB)

Advantages:

Practical: Checking storage on a computer.

A) Checking Disk Storage.

- 1) Open file explorer.
- 2) Right-click a drive.
- 3) Select properties.
- 4) See (total size, used spaces, free space)

B) Checking File Organization:

- 1) File size vs disk space used.
- 2) Fragmentation.
- 3) Cluster size.
- 4) Allocation using disk management tools.

Windows steps:

- 1) Press Window + R.
- 2) Type diskmgmt.msc
- 3) View partitions and volumes